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MAZES without minotaurs: Herbert Simon and the sciences of the artificial

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Summary In this essay, I remember Herbert Simon as author of *Sciences of the Artificial* and my thesis advisor and collaborator. Even for a lifetime of deep insights into human behavior and its role in shaping the world we live in, Simon's insight into the need for a whole new class of sciences is an astonishing one. The artificial is essential for entrepreneurship and for developing an organization-centric view of markets and economies. Published by Elsevier Ltd.

The week of June 18, 1996 was the first week of Herbert Simon's 81st year. When I asked him what he would like to have done differently if he were to live his life again, he demurred, "Not much" and then went on to muse, "perhaps, I would like to have begun earlier on challenging the centrality of markets in economics, pushing the field to start with organizations and move outward to market relations rather than the other way around." One reason for the musing could have been that the outline of his polemic on the subject had been published just a few years earlier (Simon, 1991c, p. 34) and he was continuing to work on its elaboration in at least two more talks that came after his 80th birthday (Simon, 1997b, 2000). Yet Simon's understanding of organizations began as early as his dissertation in Political Science at the University of Chicago (Simon, 1945) and remained with him through an intellectual lifetime that spanned over six decades, hundreds of articles, many of them co-authored with students and collaborators, and dozens of books, talks and interviews.

It is not easy to provide a brief biography of such a man. Nor to summarize his works in a few pages in ways that tease out his influence on management and organization science today and going forward into the future. In formulating

the ensuing account, I used several published sources including Crowther-Heyck's (2005) book on the subject and the sources that book draws upon to triangulate my conversations with him. I also scanned a number of short pieces readily available online and reread Simon's own telling of the first seventy four years of his life-story (Simon, 1991b). Finally, I found that the collection of essays by Simon's friends and colleagues in tribute to his memory provided texture and nuance to the published biographies (Augier & March, 2004).

Brief biography

Explaining the title of his autobiography, *Models of My Life*, Simon wrote:

"But there is a further reason for using the plural. It is a denial – a denial that a life, at least my life, has a central theme, a unifying thread running through it. True, there are themes (again the plural), some of the threads brighter or thicker or stronger than others. Perhaps clearest is the theme of the scientist and teacher, carrying on his persistent heuristic search, seeking the Holy Grail of truth about human decision making. In my case, even that thread is woven of finer strands: the political

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scientist, the organization theorist, the economist, the management scientist, the computer scientist, the psychologist, the philosopher of science.” (Simon, 1991b, p. xviii)

To the scientist-teacher, he then added the private person, the university politician and science politician as the four panels of the tetraptych that was his life. The multiple strands that made up the thread of his scientific life is important to understanding Simon’s contribution to organization science. In rereading the list at the end of the quote above, I could not but wonder if he wrote them down in the order in which he valued each. A similar wonderment also emerged as an evocative refrain during his memorial ceremony where each eulogist sought to claim his own discipline as somehow primary in Simon’s life and work. John R. Anderson gave explicit voice to the wonderment as follows:

“I am reminded of the story Herb told about how he amused himself as a young man when he used to travel by train across America. He would start up a conversation with some fellow traveler, inquire as to their profession, and carefully listen as they described what they did. Then when he was confident he understood the language of that profession, he would reveal to the traveler that he too shared that profession and then would try to maintain that deception for the rest of the trip. Herb must have sustained that deception for the 25 years I knew him for I am firmly convinced that he is a psychologist and that the claims of the other fields to him are simply false.”¹

After almost six years of conversations on the subject, I feel a similar temptation to cast him as an organization scientist with a particular interest in how *new* – i.e. entrepreneurial – organizations come to be. His autobiography provides some fodder for the tempting folly through one of the most profound and transformative experiences of Simon’s youth² – an entrepreneurial (mis?)adventure called Rockmarsh:

“The real adventure of my life was Rockmarsh. It was real to me, perhaps surreal, for the same reason that Don Quixote was real to Cervantes – more real than the Battle of Lepanto that cost him an arm.” (Simon, 1991b, p. 29)

Born the younger of two sons to Arthur, an engineer-turned-patent attorney and Edna, a piano teacher and homemaker – both of German-Jewish heritage, young Herbert Simon spent most of his early life in the American state of Wisconsin. Simon’s Rockmarsh adventure took place during the summers of his sixteenth through twenty-first years. The “entrepreneur” behind Rockmarsh was Maurice Davis, veteran of World War I and promoter of a scheme to transform marshlands into highly profitable pastures for dairy cattle by planting them in reed canary grass. Davis convinced Simon’s father to invest in the venture and Simon ended up working as Davis’ right hand man (or boy) in getting the enterprise going. The history of the venture had

several of the recurring drama of new ventures everywhere and everywhen – technological and logistical surprises, unpredictable market responses and of course, far-from-rational human beings capable of glory and pathos all at once. Simon avers:

“Such interpersonal skills and understanding of people outside academe and the professions as I possess owe more to Rockmarsh than to any other single experience I have had.” (Simon, 1991b, p. 34)

The venture did not end well. Davis, who was manic-depressive, committed suicide. And the cattle simply would NOT eat reed canary grass:

“They would eat anything but reed canary grass, and broke down every fence we constructed, including barbed and electric, to get out of the pasture.” (Simon, 1991b, p. 35)

Simon carried this formative experience into the University of Chicago where he earned both a bachelor’s degree (in 1936) and a PhD in political science (in 1943) while studying under and working with a pantheon of social scientists and philosophers such as Henry Simons, Nicholas Rashevsky, Henry Schultz, Rudolf Carnap, Charles Merriam, Harold Lasswell and Clarence Ridley. What Simon took away from Chicago was not merely the content of courses or even the intellectual challenges the high-octane community provided him with, but a sense of how to *do* social science and the *social* aspects of doing it well. In his own words:

“As for myself, there could have been no better school than the Political Science Department at Chicago to teach me about the march of ideas, and how the interplay of scientific research with the social organization of the disciplines determines its direction and pace. It helped me understand that new ideas do not fly solely on their own wings; the scientist is a communicator as well as a discoverer – sometimes even a missionary.” (Simon, 1991b, p. 63)

From biography to dissertation: foundational concepts of organization science

Simon’s dissertation, published with the title *Administrative Behavior* has been hailed as one of the most influential books in the social sciences and prescribed as required reading in a variety of doctoral courses even today. Yet it has not been without controversy through its four editions – with critics questioning the necessity or even possibility of studying administration as a science (Harmon, 1959; Storing, 1962) let alone the wisdom of doing so for fear it drive the field down narrow or even unprogressive paths (Argyris, 1973a,b; Waldo, 1952a,b). Yet it was the intensity of the empirical experience on which the book was based combined with the great conceptual care it took to create a precise “vocabulary” with which to describe that empirical experience that formed the foundation of a *science* of organizations as we know it today.

For Simon, the term *organizations* did not merely refer to an alternate form of coordination to that of markets. In fact, in later years, he expended considerable energy and

¹ A tribute to Herbert Simon. All remembrances. Available at: <http://www.cs.cmu.edu/simon/all.html>.

² Explicitly acknowledged as such in the footnote at the bottom of page 28 in *Models of My Life*.

thought in arguing otherwise. Through the Raffaele Mattioli lecture, for example, he vigorously urged economists to flesh out the notion of organizations, criticizing extant conceptions as being almost entirely “skin”, in blatant contrast to the “full-bodied” firms that populate the real world of business (Simon, 1997b, pp. 35–36). Even though he explicitly enjoined the economists in this matter only later in his career, he had made essentially the same argument to political scientists at the beginning of his career. In the preface to the first edition of *Administrative Behavior*, he stated:

“Among the studies of administrative organizations that I have read, few have caught and set down in words the real flesh and bones of an organization.” (Simon, 1997a, p. xi)

Also, in Simon (1946, pp. 62–63):

“What is a scientifically relevant description of an organization? It is a description that, so far as possible, designates for each person in the organization what decisions that person makes and the influences to which he is subject in making each of these decisions. Current descriptions of administrative organizations fall far short of this standard. For the most part, they confine themselves to the allocation of functions and the formal structure of authority. They give little attention to the other types of organizational influence or to the system of communication. (...) In sum, a description of administrative organizations in terms almost exclusively of functions and lines of authority is completely inadequate for purposes of administrative analysis.”

In the fourth edition of *Administrative Behavior*, he further clarified the above definition of organizations as follows:

“In this book, the term *organization* refers to the pattern of communications and relations among a group of human beings, including the processes for making and implementing decisions. This pattern provides to organization members much of the information and many of the assumptions, goals and attitudes that enter into their decision, and provides also a set of stable and comprehensible expectations as to what the other members of the group are doing and how they will react to what one says and does.” (Simon, 1997a, pp. 18–19)

This definition brings communication, relations and processes front and center as the key conceptual “flesh” of organizations and also stakes out decision-making as their “heart” – implying that any science of organizations must consider decisions as the primary unit of analysis. This focus on decision making and problem solving has remained at the heart of much of Simon’s work, whether encapsulated in formal modeling and laboratory experimentation, or wrapped up in historical and philosophical methodologies, or his clear ongoing favorite, “common observation and experience.” (Simon, 1991b, p. 59)

Yet in recent times, most management and organization scientists seem to be moving away from decisions to other foci such as routines, transactions and capabilities with an overarching emphasis on adaptation to external changes (Gavetti, Levinthal, & Ocasio, 2007). The increasing rate of

technological change explains only a small part of this shift. Another part may consist in the reality of a globalized world threatened by climate change and financial collapse at least in part triggered by the (mis)behavior of large organizations. (Walsh, Meyer, & Schoonhoven, 2006). Furthermore, purely in terms of the “social organization of disciplines” that Simon referred to earlier, the tides have been shifting away from psychological to sociological approaches due to more scholars trained in sociology coming into business schools and training generations of doctoral students who then go on to edit journals and foster schools of thought that draw upon the vocabulary of their disciplinary mother tongue.

On the sociology and/or psychology of organizations

As Simon’s experience at the University of Chicago taught him, the question of disciplinary influence is not purely a matter of scientific propriety, it is also a matter of the seeding, nurturing and propagation of particular schools of thought within specific communities of practitioners. As Gavetti et al. (2007) chronicle its history, the Carnegie School founded by Simon, March, Cyert and others urged a focus on decisions within a theoretical edifice that consisted of empirically observed behavioral and organizational concepts such as bounded rationality, satisficing based on changing aspiration levels, authority relations in hierarchical decision structures, information flows through political coalitions and “the delicate conversion of conflict into cooperation” (March & Simon, 1993, p. 2). Also, the School spawned several heterodoxies such as Jim March’s sparkling pieces on organizational learning and Nelson and Winter’s pioneering work in evolutionary economics (Nelson & Winter, 1982). Both paid considerable attention to the dynamics within and outside the organization.

Through all these, however, the Carnegie School refused to avow its loyalties to any one particular discipline. In the meanwhile, a growing contingent of economists began taking psychological approaches more seriously, and management and organizational researchers moved away from the decision-centered view of organizations to more “structural” explanations that are more explicitly amenable to a sociological worldview.

Simon was not unaware of this movement. In his commentaries on later editions of *Administrative Behavior* as well as through dialog pieces and full-fledged articles during his later years, he spoke up for the centrality of decisions and decision-making without having to follow any one disciplinary orthodoxy. Of course, those who knew him well would readily assert that if there was one thing you could count on as an invariant through the multiple strands that wove together the tapestry that was Simon’s oeuvre, it was his rejection of disciplinary orthodoxy. He moved easily between disciplines, sometimes borrowing from one to challenge the orthodoxy of another, sometimes transforming a discipline from within and at others inventing a whole new discipline if needed, building its foundations from scratch. In each of these projects, his stance was committed and determined yet playful and irreverent. Let us take just two passing examples of this unmistakable style and then examine a third in more depth in the next section.

In crafting compelling arguments whether for or against an idea he cared about, Simon was prone to use analogies from the natural sciences. The history and psychology of science was a continuing phenomenon of interest throughout his intellectual career. He told me once not to worry too much about the philosophy of science³ and to examine instead how real scientists actually *did* science. It was no surprise then, that when confronted with the issue of whether organizations should be analyzed using a psychological or sociological lens, he responded with:

“The question is a bit odd: it is like asking whether molecular biology is biology or chemistry. The correct answer in either case is ‘both’.” (Simon, 1997a, p. 24)

And then went on to dismiss the relevance of disciplinary labels altogether:

“Readers can decide for themselves, while they continue through the pages of this book, whether they are reading ‘psychology’ or ‘sociology’ or they can decide that it doesn’t matter. I confess I hold the latter view.” (Simon, 1997b, p. 25)

Similarly when asked to discuss organizational learning at a conference in honor of Jim March, he began by almost defining the phenomenon out of existence, yet regrouped with a series of characteristically simple yet insightful observations about it:

“All learning takes place inside individual human heads; an organization learns in only two ways: (a) by the learning of its members, or (b) by ingesting new members who have knowledge the organization didn’t previously have. But what is stored in any one head in an organization may not be unrelated to what is stored in other heads; and the relation between those two (and other) stores may have a great bearing on how the organization operates.” (Simon, 1991a, p. 125)

Later in the same discussion, Simon conscripted the canonical concept of “roles” from sociology to facilitate a more textured analysis of one of the central concepts of his own theorizing about organizations, namely, decision premises:

“The point has perhaps not been emphasized in the sociological literature as often as it should be that a role is not a system of prescribed behaviors but a system of prescribed decision premises.” (Simon, 1991a, p. 126)

Whereas both these examples show Simon’s instinctive irreverence for any kind of disciplinary orthodoxy, they also illustrate how adroitly he could put any discipline to work for the phenomena he cared about and was committed to understand. For a deeper look at this fascinating *modus operandi*, we should turn to *The Sciences of The Artificial*. Here we can witness the sheer effortless ease with which Simon could weave together masterful insights from multiple disciplines to kickstart an entirely new class of sciences.

The design of new organizations or organization as artifact

It is customary to think of two classes of sciences — natural and social. And it is routine to classify fields such as management and organizational studies under the social sciences. There are at least two major problems with this classification. First, it creates the illusion that the two categories exhaust all phenomena of interest to scientists; and second, it obscures one of the most fascinating aspects of the phenomena we file away under the rubric of social sciences — namely the fact that they have to do with what *can* be rather than with what is or what ought to be. Simon expressed his insight as follows:

“Finally, I thought I began to see in the problem of artificiality an explanation of the difficulty that has been experienced in filling engineering and other professions with empirical and theoretical substance distinct from the substance of their supporting sciences. Engineering, medicine, business, architecture, and painting are concerned not with the necessary but with the contingent — not with how things are but with how they might be — in short, with design.” (Simon, 1981, p. xi)

More generally, artifactual phenomena embody teleology⁴. That does not mean, however, that we cannot study them scientifically. Nor does it mean that we can afford to ignore the laws of nature, and/or relevant theories from the social sciences, as they apply to artifacts. Simon’s insight, forged through decades of trying to deepen our understanding of organizational and other phenomena that involve motivations, aspirations, purposes — often ambiguous yet operating within contexts of multiple uncertainties — not only foreshadowed our recent angst about the prescriptive relevance of our scholarship (McGahan, 2007), but also pointed out the possibility of constructing an entirely new class of sciences where such relevance is stitched into its methods. The task he is pointing to here is no doubt massive and daunting, but it is at its core a problem that organizational scientists should not shy away from. Simon’s own words provide us with actionable cues as well as inspiration:

“Thus we are faced with a problem of devising a professional school that can attain two objectives simultaneously: education in both artificial and natural science at a high intellectual level. This too is a problem of design — organizational design.

The kernel of this problem lies in the phrase “artificial science.” In my previous chapters I have shown that a science of artificial phenomena is always in imminent danger of dissolving and vanishing. The peculiar properties of the artifact lie on the thin interface between the natural laws within it and the natural laws without. What can we say about it? What is there to study besides the boundary sciences — those that govern the means and the task environment? (...) The proper study of those

³ I had been busy reading up on the debate between Carnap and Popper at the time and the resulting flirtation with metaphysics was interfering with my research design.

⁴ In the context of entrepreneurship, the science of the artificial is relabelled the science of the artifactual, which allows us to ask design-oriented questions rather than “why” questions (*Note from the guest editors*).

who are concerned with the artificial is... the process of design itself. The professional schools will reassume their professional responsibilities just to the degree that they can discover a science of design, a body of intellectually tough, analytic, partly formalizable, partly empirical, teachable doctrine about the design process.” (Simon, 1981, pp. 131–132)

The book is filled with examples of phenomena that are products of human artifice, yet amenable to scientific study that goes beyond “applying” theories from the natural and social sciences. These include organizations and markets, computers, bridges, robots and other products of engineering, languages, even society itself. To that list, I would add entrepreneurship. I had the good fortune to work with Simon for over five years on formulating entrepreneurship as a science of the artificial. I have written about that elsewhere (Augier & Sarasvathy, 2004; Sarasvathy, 2003; Sarasvathy, Dew, Read, & Wiltbank, 2008; Sarasvathy, 2008). Here I would like to use entrepreneurship simply as a prototypical example of an artifactual science.

Let us begin by asking the obvious pragmatic question: What difference does it make whether we study any particular phenomenon as natural, social or artifactual? Natural sciences eschew teleological explanations and tend to focus almost entirely on material and efficient causes. Social sciences tend to emulate the natural sciences in the realm of humans – again by looking for material and efficient causes to explain human behavior or the behavior of social systems. Teleological explanations have to do with attributions of purpose to occurrences. For example, let us say we observe that the grass on our front yard is wet. Natural science would seek to model the probability that it rained (based on local weather patterns for example); social sciences would try to predict the probability that someone turned on the hose (based on the behavioral patterns of the homeowner for example); and a science of the artificial would tackle the issue of whether and when to set the sprinklers to go on and off and also help devise entirely new ways of getting the grass wet or not, depending on the homeowner’s wishes, but within the constraints of local weather conditions and the bylaws of the homeowners’ association. Note that in the case of the artifactual, we do not deal only with extant phenomena (the grass is wet), but also with future possibilities (to water the lawn or not and how and how else). And even when we take natural laws or social norms as given, we still have plenty of room for choice and for creative choice at that.

Now let us take a brief look at a more complex phenomenon: entrepreneurship. We can choose to explain the existence of firms through facilitating economic institutions such as property rights and the availability of credit (using a social science perspective); we can also seek to explain it in terms of technological possibilities, demographics and other factors governed by natural laws. Simon’s vision in *The Sciences of The Artificial* suggests we can also try to understand and teach scientific ways to *design* firms and even industries and institutions. For example, we can consciously design experiments not only to prove or disprove some of our theoretically driven hypotheses, but we can empirically test key design principles to build future institutions by actually building them – through simulations, thought experiments and even artificial field experiments.

As Simon rightly pointed out, people inside and outside the official halls of science are already *doing* artifactual sciences. But in many ways, the sciences of the artificial are for the most part still in the domain of amateur artisans than trained scientists – we are back in Francis Bacon’s time rather than in Alfred North Whitehead’s. This may be cause for excitement rather than lament for those of us willing to engage in building sciences of the artificial – at least that is the spirit in which I will tackle the concluding section of this essay.

Implications for the future of organization sciences

One can imagine a wide variety of issues, problems and challenges that face scholars in business management and organizations, each of which can be solved by insights in Herbert Simon’s vast body of works. I have highlighted only three elements from his oeuvre: (1) A decision-centric view of organizations; (2) An organization-centric view of market economies; and (3) An artifactual view of both organizations and markets. Each of these suggest possibilities for future research that can tackle some of the toughest problems facing us not only as a community of scientists, but as human beings, as a single species across a lonely planet that has somehow managed to survive and may be even come of age.

Let us consider one such problem: the recent global economic crisis engendered by the fall of big banks such as Lehman. This crisis has been attributed to a variety of causes from unbridled capitalistic structures to immoral greed and stupidity of investment bankers and homeowners, not to mention the simple-minded “genius” of mathematicians devising exotic financial instruments such as the credit default swaps that allowed the greedy and stupid to unhook their actions from the constraints of reality. Despite the sweeping range of possible explanations, a large consensus does exist for at least one proximate cause – namely the decision by banks to begin reselling and securitizing mortgages. Once upon a time, the story goes, local banks used to take in deposits from local people and lend them out to local homebuyers. The spread was small and for the most part everyone lived and slept well and hardly anyone got enormously wealthy. And then one day, banks decided to resell these mortgages to other banks who then bundled them up and resold them to others and eventually entirely new “bundles” and new types of “banks” were invented until the entire system became so far unhooked from reality that even Lewis Carroll’s Queen could not swallow it, before or after breakfast (Johnson & Kwak, 2010).

A decision-centric view of organizations would allow us to tackle this proximate cause not only as individual moral failure or societal moral corruption, but as a problem of concrete decision premises, processes, relationships and information flows at the organizational level. The crisis is not only a matter of study for economists and financial theorists or even for philosophers and politicians – it has substantial roots and remedies in organizational design.

There is a credible role here for an organization-centric view of markets as well. The financial crisis is a classic case of the failure of organizations leading to the failure of markets. And ought to be studied as such. Moreover, we would

be overlooking an important recourse in rebuilding financial markets if we see it merely as a task for policymakers and activists. It is a task for organization scientists and entrepreneurs (designers of new organizations) as well. For example, entrepreneurs around the world are questioning the wisdom of separating for-profit and nonprofit organizations and inventing hybrids that seek to put business to work in solving social problems. Mercy Corps, a nonprofit organization headquartered in the United States recently purchased a for-profit bank in Indonesia. Such an organizational innovation rearranges not only the relations between for-profit and nonprofit motivations that impact decision making within its own skin, but causes ripples outward into the very notion of "market" at the heart of most market economies. At the very least, the case of Mercy Corps is challenging American legal and fiscal institutions that prohibit nonprofits from earning profits. Surely, phenomena such as this should be a matter of avid interest to organizational and management researchers everywhere? Are we to wait until the ad hoc new institutions and organizational structures locally formulated by entities such as Mercy Corps or Grameen Bank become a widespread mainstream phenomenon of interest to scholarship? Or should we get ahead of the "evolution" of this phenomenon and treat it as an instance worthy of a science of the artificial?

There is something inherently optimistic at the heart of the term *Sciences of the Artificial*. At its most grandiose, it suggests we can design a better world. But even at its most mundane, it prods us to roll up our sleeves and tackle the problems we see as worth tackling — and argues there is always room for us to solve them even within the constraints of natural and social forces that may be outside our direct control. In my own work, I take inspiration from Simon about being bullish about the future of our field and our species. And in terms of *doing science* — natural, social or artificial — I have learned not to apologize for intellectual curiosity mixing with what Einstein called "einfuhlung" for the phenomenon — for as one of Simon's favorite authors put it:

"But what man can content himself with seeking out proofs for a thing that not even he himself believes in, or whose teaching he cares naught for?" (Borges, *Three Versions of Judas*, 1962, pp. 151–157).

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